

Molecular Basis of Inheritance - 100 MCQs

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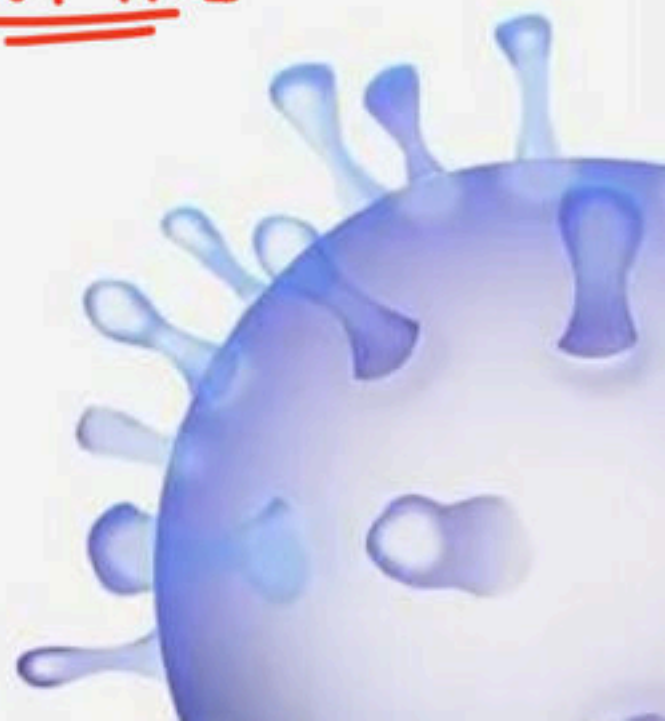
NCERT Nichod : Biology Practice Series NEET 2025



Good morning
↓
audible?
↓
Ready?

Molecular Basis of Inheritance

✓ - 80 MKS



BIOLOGYLIVE

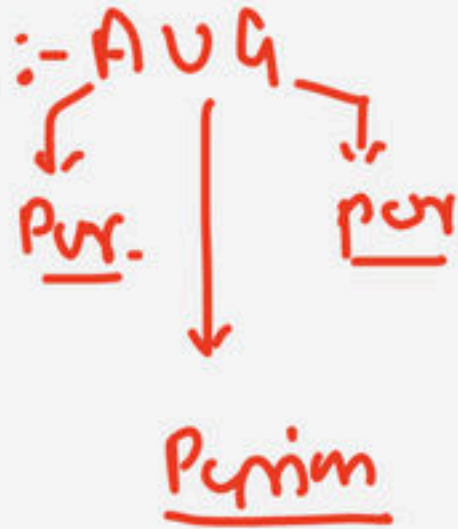
1. DNA polymorphism forms the basis of :

- (1) DNA finger printing
- ✓ (2) Both genetic mapping and DNA finger printing
- (3) Translation
- (4) Genetic mapping



2. The ratio between purines and pyrimidines in the starting codon

v. imp



2 : 1

✓ (1) 2 : 1

(2) 1 : 1

(3) 1 : 2

(4) 3 : 1



3. The cellular factory / Organelle responsible for the synthesis of proteins is

✓(1) Ribosome ↩

(2) Nucleus

(3) Golgi body

(4) Centriole



4. A coding sequence made of alternating C and U bases would form a polypeptide having —



- (1) Either leu or ser residue
- ☒ (2) Alterating leu and ser residues
- (3) Only ser residues
- (4) Only leu residues



5. Presence of exons and introns is seen in

(1) Secondary transcript

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✓ (2) Primary transcript

(3) t RNA

(4) snRNA



6. DNA-dependent RNA polymerase also catalyse the polymerisation in only one direction,

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(1) $3' \rightarrow 3'$

(2) $5' \rightarrow 5'$

✓(3) $5' \rightarrow 3'$

(4) $3' \rightarrow 5'$



7. In eukaryotes, RNA polymerase III catalyses the synthesis of :-

- ✓ (1) 5 srRNA, tRNA & SnRNA
- (2) mRNA, HnRNA & SnRNA
- (3) 28 S rRNA, 18 S rRNA & 5 S rRNA
- (4) All types of rRNA & tRNA

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8. Transcription and translation can be couple in:- Prokary.

(1) Yeast

(2) Paramecium

✓(3) E.Coli

(4) Amoeba



9. Discontinuously synthesized strands of DNA adjacent to ^{5' → 3'} DNA template are joined together by _____.

V. Imp

(1) 5 → 3, DNA endonuclease ^{5' - 3' =} discont.

(2) 3' → 5' DNA Aase

✓ (3) ["] 5' → 3' DNA ligase

(4) 3' → 5', DNA ligase



10. Transcription is intiated by :-

- (1) Only σ (sigma) factor
- (2) Only DNA dependent RNA polymerase
- (3) Only ρ (Rho) factor
- ✓ (4) RNA polymerase when it associates transiently with σ (sigma) factor



11. In a crime investigation, the investigating officer collects different biological samples from the crime spot for DNA fingerprinting analysis. Which of the following samples is not helpful in this analysis?

(1) Skin Shreds

✓(2) Erythrocytes

(3) Semen sample

(4) Hair follicle



12. What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

(1) Transcription of rRNAs (28S, 18S and 5.8S)

✓✓ (2) Transcription of tRNA, 5S rRNA and snRNA

(3) Transcription of precursor of mRNA

(4) Transcription of only snRNAs



13 Genetic material of bacteriophage $\phi \times 174$ and it has
_____ nucleotides

↓
ssDNA

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- ✓✓ (1) ss DNA, 5386
- (2) ss RNA, 5386
- (3) ds DNA, 5368
- (4) ss DNA, 5368



14. In a 3.2 Kbp long piece of DNA, 820 adenine bases were found. What would be the number of cytosine bases?

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(1) 1560

✓ (2) 780

(3) 1480

(4) 740

$$3.2 \text{ Kbp} = \underline{3200 \text{ bp}}$$

$$\text{If, } A = 820; T = 820$$

$$\text{Purin} = \text{Pyrim} \begin{matrix} \nearrow A = T \\ \searrow G = C \end{matrix}$$

$$820 + 820 = 1640$$

$$3200 - 1640 = \underline{\underline{1560}}$$

$$G + C = 1560$$

$$C = \frac{1560}{2} = \underline{\underline{780}}$$



15. If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as : 202.

- (1) Gene mapping
- (2) Expressed sequence tags
- (3) Bioinformatics
- ✓ (4) Sequence annotation ✓



16. Select the correct option

i. z gene codes → beta-galactosidase (β -gal) ✓

ii. y gene codes → permease ✓

iii. a gene → transacetylase ✓

iv. i gene → repressor of the lac operon ✓

(1) Only i is correct

(2) Only ii,iii is correct

(3) Only i, ii, iii is correct

✓ (4) All are correct



17. If the sequence of bases in DNA is AATCCATG, then the sequence of bases in its transcript will be :-

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UUAGGUAC

☒ (1) UUAGGUAC

(2) AAUCGAAU

(3) UUAGCUUA

(4) AAUCGAUG



18. The radioactive probe used by Taylor to detect distribution of newly synthesized DNA in chromosomes:

- ✓ (1) Thymidine ↪
- (2) Adenosine
- (3) Guanidine
- (4) Cytosine



19 ^{V.I.M} The year 1953 was known for the Double helical model of DNA given by____
, also for discovery of ribosome by____, also for proposing the semi
conservative DNA replication by____ and also for the experiment done on
chemical evolution by____

- ☒ (1) Watson and Crick; G.Palade; Watson and Crick; Urey Miller
(2) Watson and Crick; G.Palade; Messelson and Stahl; Stanley Miller
(3) Watson and Crick; De Duve; Messelson and Stahl; Stanley Miller
(4) Watson and Crick; G.Palade; Watson and Crick; Stanley Miller



20. UUU, UUC code for which amino acids ?

- (1) Alanine, phenylalanine respectively
- (2) Phenylalanine, alanine respectively
- (3) Phenylalanine, methionine respectively
- ✓✓ (4) Phenylalanine, phenylalanine respectively



21. For transcription the requirements are :-

i. DNA ✓

ii. DNA dependent RNA polymerase ✓

iii. Nucleoside triphosphate ✓

iv. Helicase X
↳ Replication

✓ (1) i,ii,iii

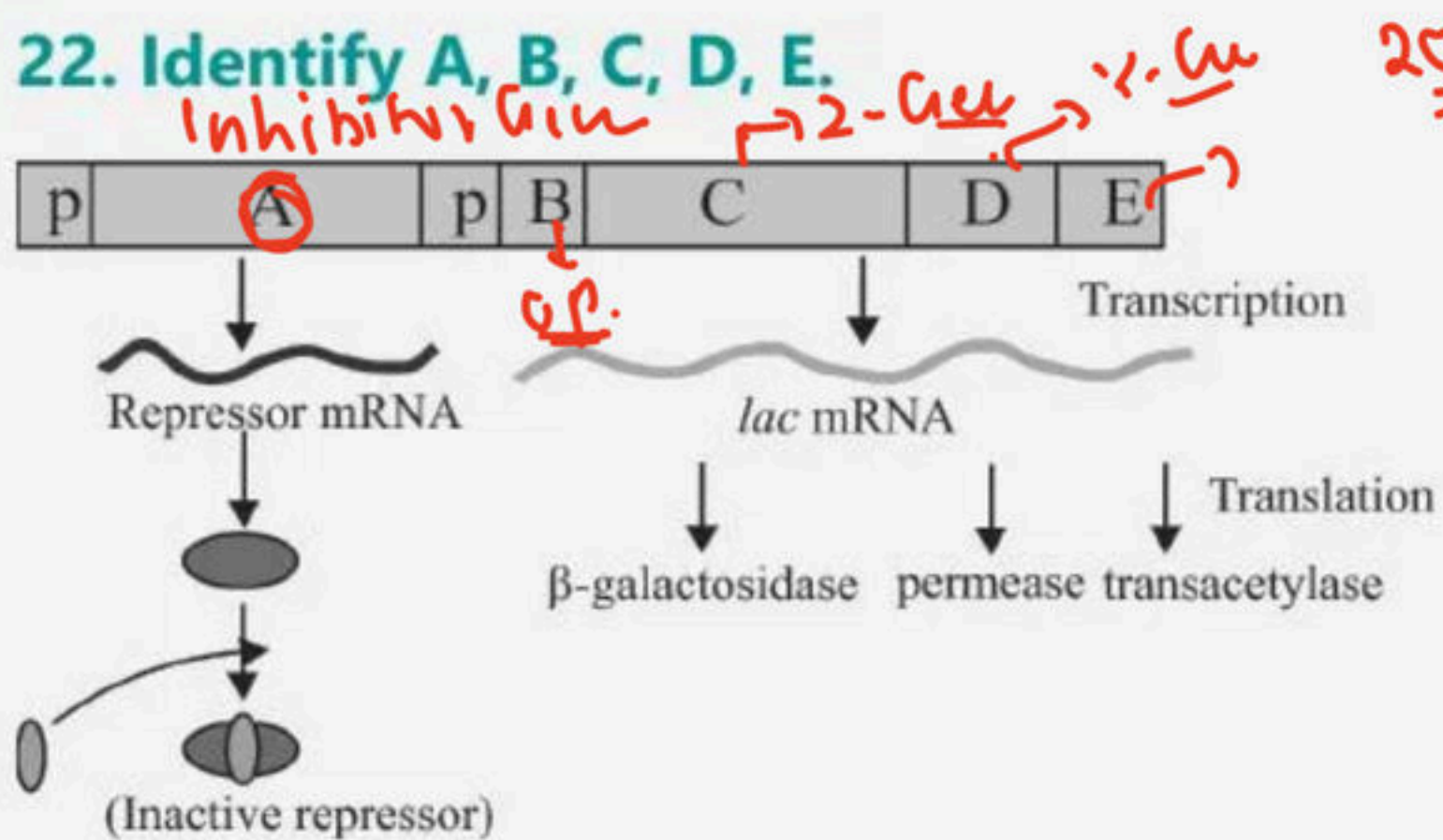
(2) i,ii only

(3) i,ii,iv

(4) All



22. Identify A, B, C, D, E.



A = Inhibitor B = Operator
 (1) C = y gene D = z gene
 E = a gene

A = inducer B = Operator
 (2) C = z gene D = y gene
 E = a gene

A = Inhibitor gene B = Operator
 (3) C = z gene D = y gene
 E = a gene

A = inducer B = Operator
 (4) C = y gene D = z gene
 E = a gene



²⁰²⁵
23. During transcription the DNA strand with 3' → 5' polarity of the structural gene always acts as a template because

- (1) Enzyme DNA dependent RNA polymerase always catalyse the polymerisation in 3' → 5' direction
- (2) Enzyme DNA dependent RNA polymerase always catalyse polymerisation in both the directions
- (3) Nucleotides of DNA strand with 5' → 3' are transferred to mRNA
- ✓ (4) Enzyme DNA dependent RNA polymerase always catalyse the polymerisation in 5' → 3' direction



24. Select the correct statement

- (1) Two strands of DNA are parallel and complimentary
- (2) Two strands of DNA are anti- parallel and noncomplimentary
- ✓ (3) Two strands are anti- parallel and complimentary ✓
- (4) All the above

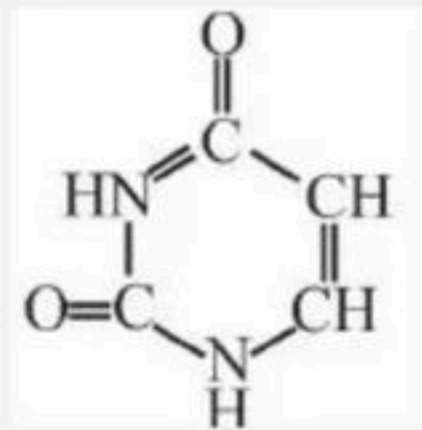


25. Haploid content of human DNA is ____

- (1) 1.65×10^9 bP
- (2) $3.1 - 3.3 \times 10^9$ bP
- (3) 4.6×10^9 bP
- ✓ (4) 3.3×10^9 bP ✓



26 Identify the structure.



(1) Cytosine

(2) Thiamine

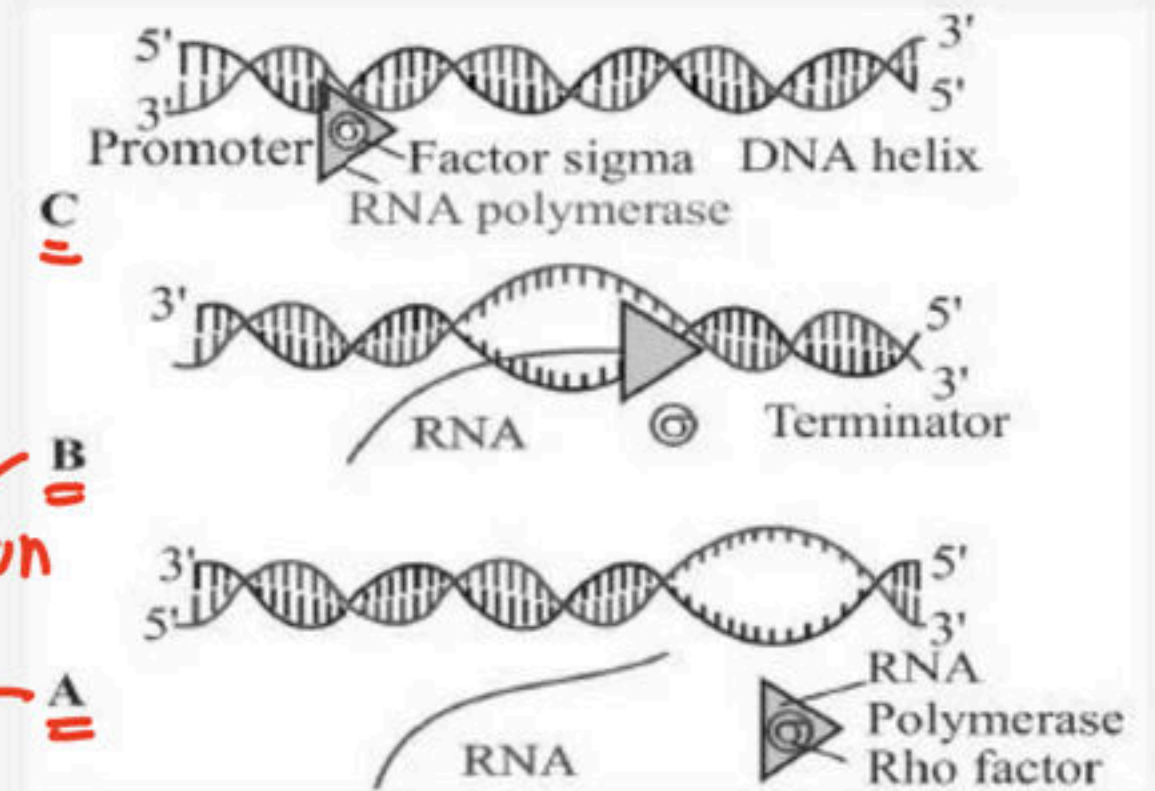
✓(3) Uracil ✓

(4) Thymine



27. In the following identify the name of the process

- (1) A - Initiation, B - Elongation, C - Termination
- ✓ (2) A - Termination, B - Elongation, C - Initiation
- (3) A - Elongation, B - Initiation, C - Termination
- (4) A - termination, B - Initiation, - Elongation

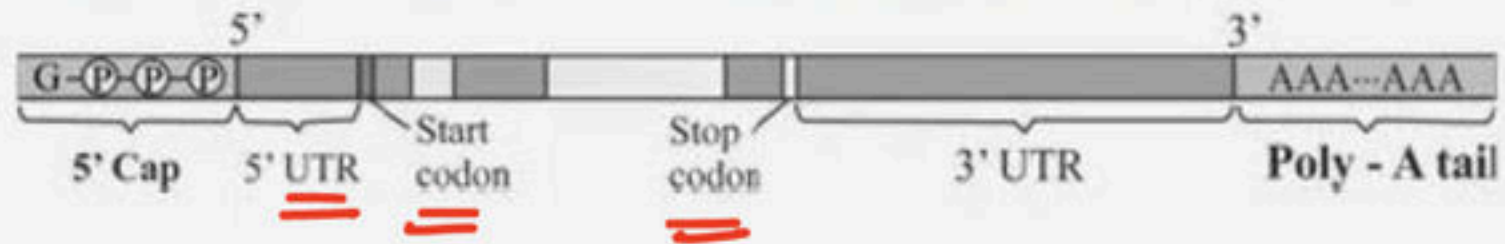


←
Elongation

↘
Termination



28. The following structure represents.



(1) r-RNA

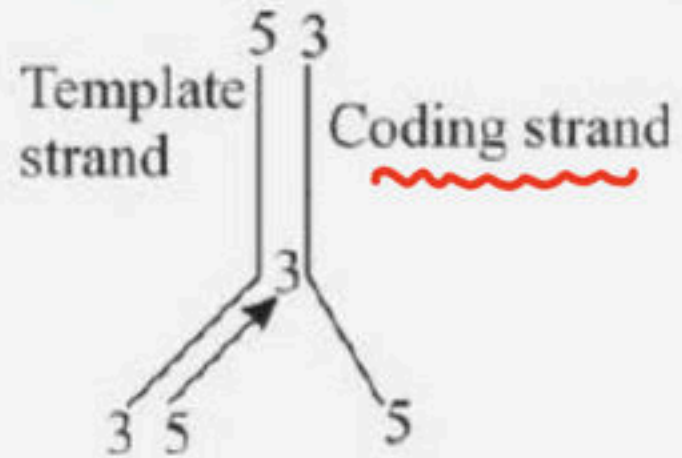
(2) hn-RNA

✓ (3) m-RNA / *Processed RNA*

(4) t-RNA



29. This Diagram indicates :-



(1) DNA replication

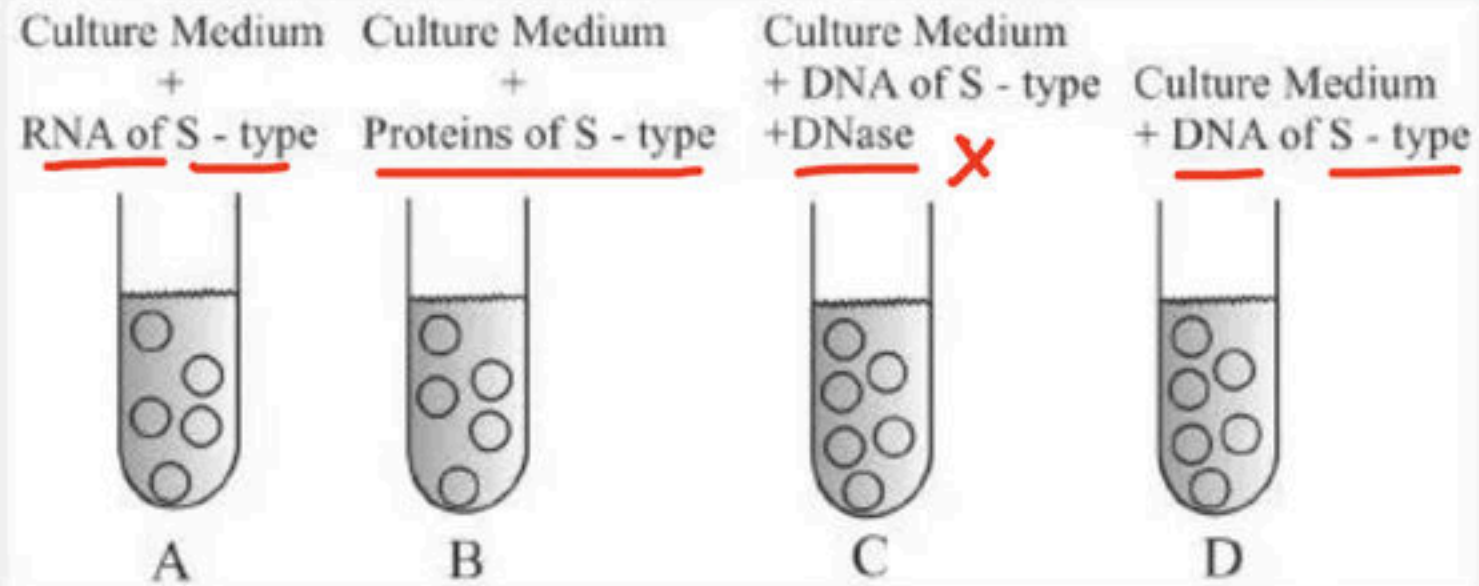
(2) Reverse transcription

✓ (3) Transcription ✓

(4) Translation



30. Mention in which test-tube did Avery et al found transformation :-



→ transformation

(1) A

(2) B

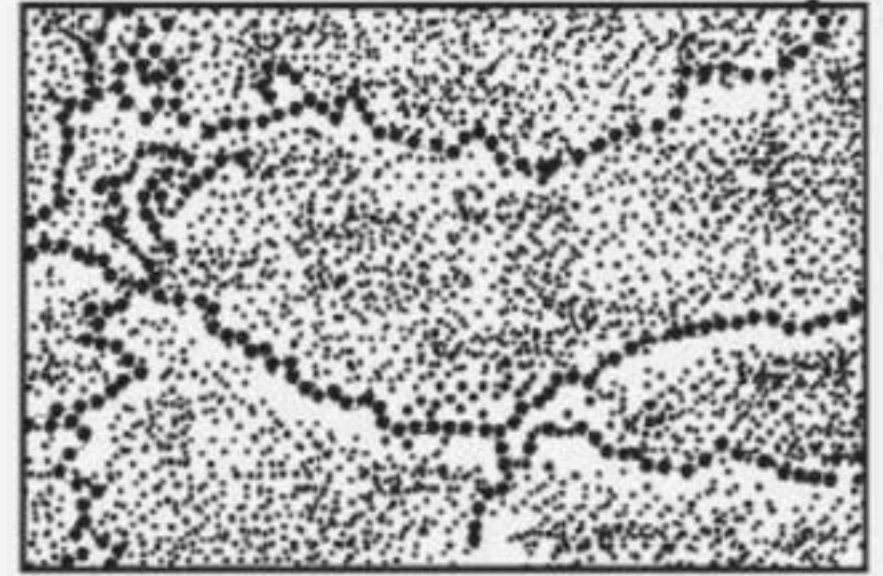
(3) C

☒ (4) D

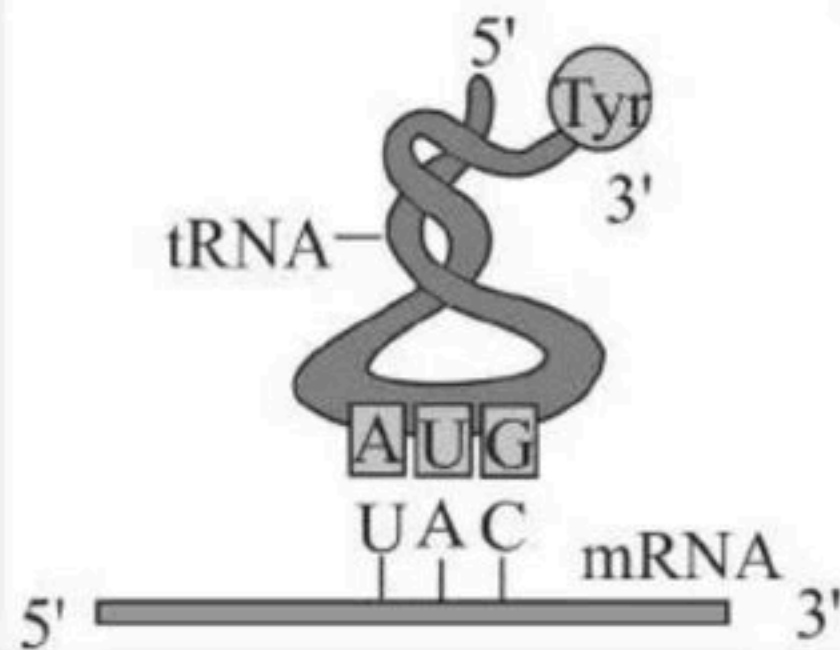


31. The following structure represents

- ✓ (1) 'Beads on string' structure of nucleosome in chromatin
- ✓ (2) 'Beads on string' under electron microscope
- (3) 'Beads on string' under compound microscope
- ✓ (4) Both 1 & 2



32. In the following diagram the direction of anticodon is?



t-RNA = 3' - 5'

(1) t-RNA -3'-3'

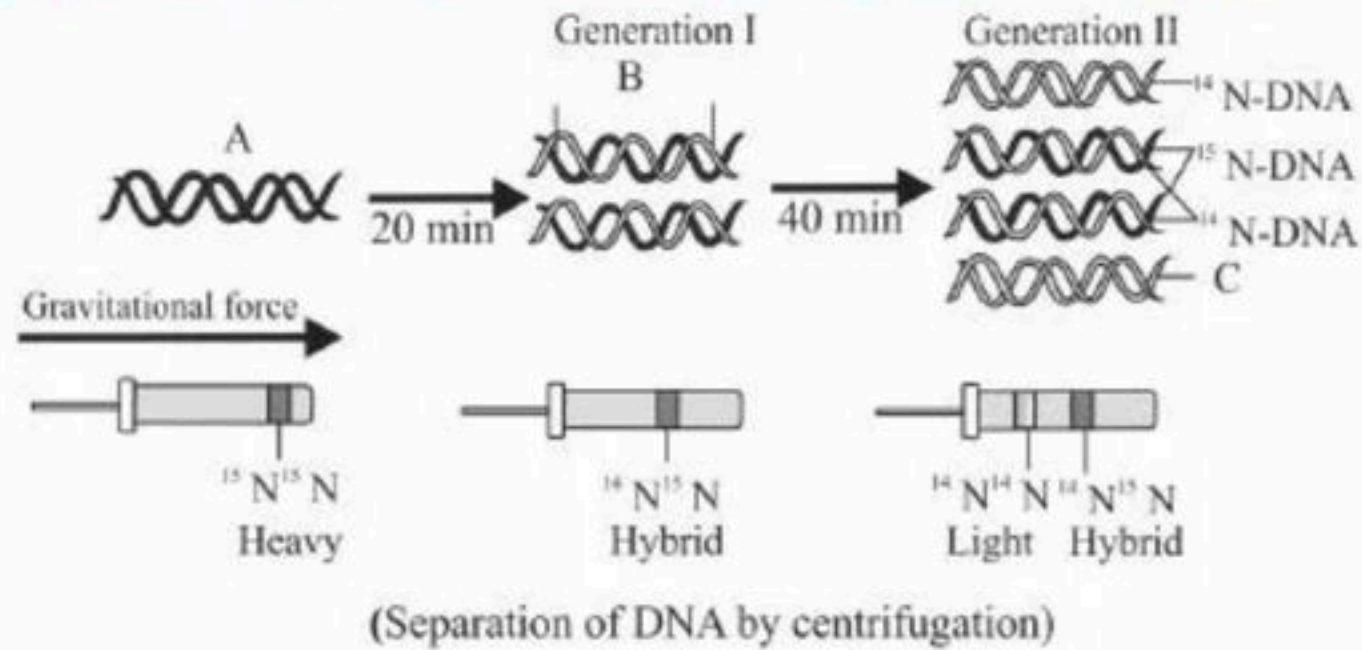
✓ (2) t-RNA-3'-5'

(3) t-RNA-5'-3'

(4) t-RNA-5'-5'



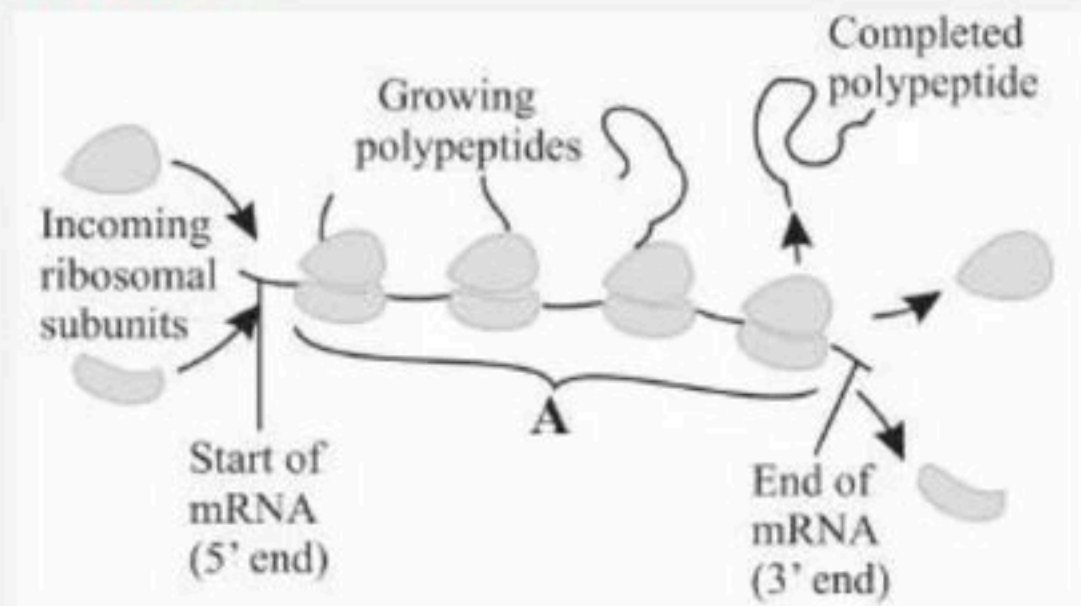
33. Which experiment is shown in the below figure?



- (1) Griffith's experiment (1928)
- (2) Hershey and Chase experiment (1952)
- ✓ (3) Meselson and Stahl experiment (1958)
- (4) Watson and Crick experiment



34. The following represents a polysome, how many m- RNA are represented here ?



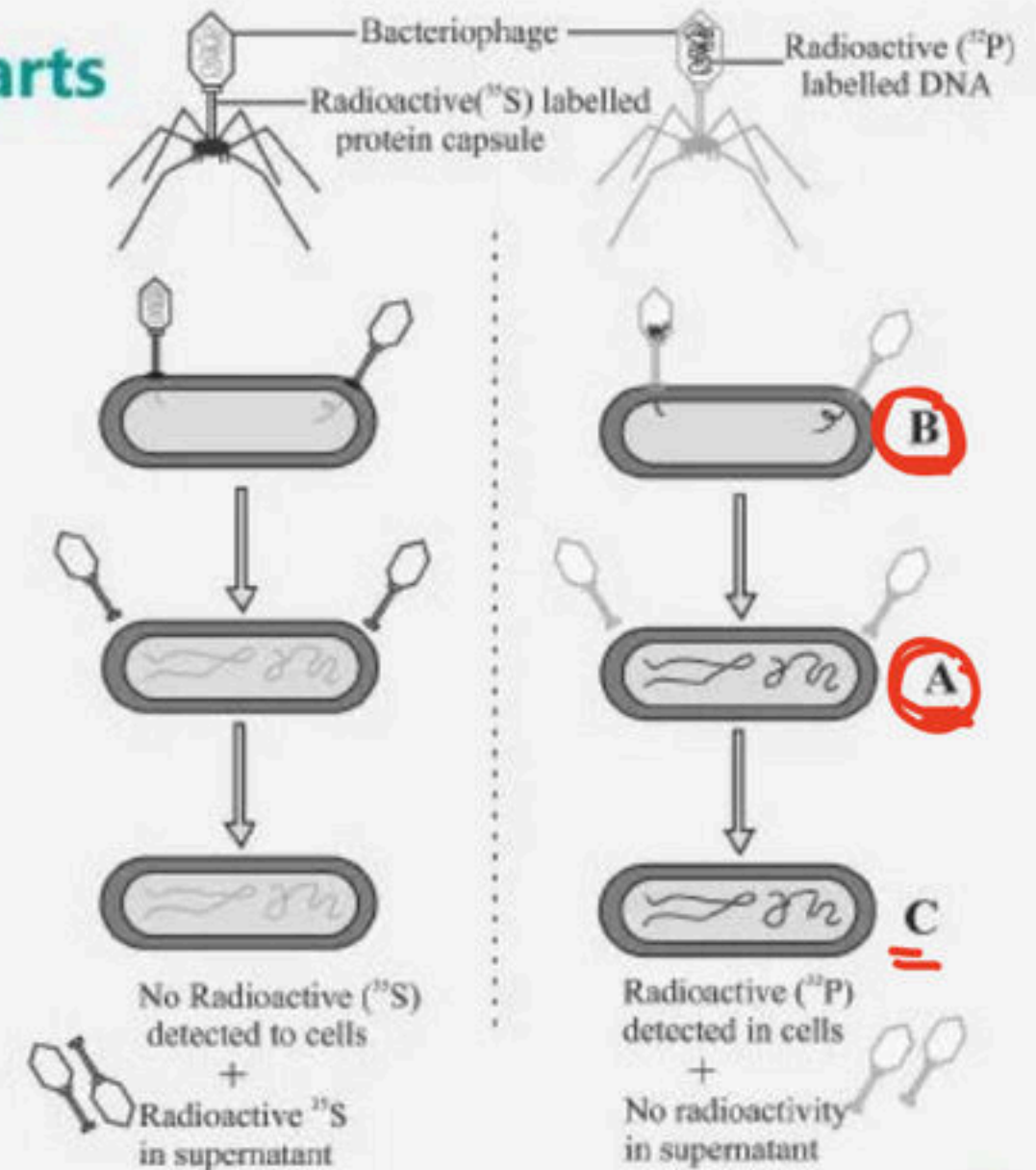
- ✓(1) Only one
- (2) As many as polypeptides are present
- (3) It can be variable
- (4) None of these



V.IMP

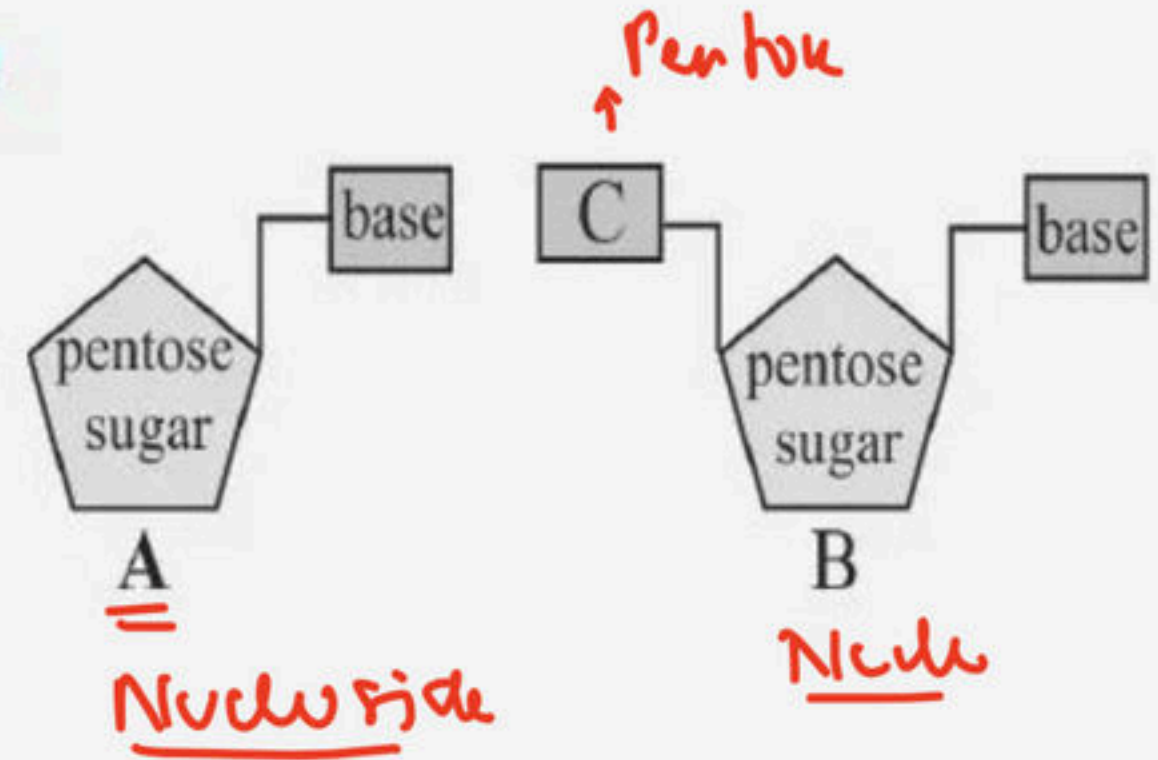
35. In the following experiment identify the labelled parts

- (1) A - Infection, B - Blending, C - Centrifugation
- (2) A - Centrifugation, B - Infection, C - Blending
- ✓ (3) A - Blending, B - Infection, C - Centrifugation
- (4) A - Centrifugation, B - Blending, C - Infection

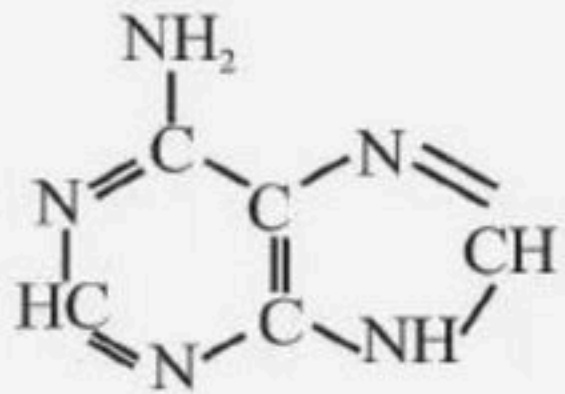


36. Choose the correct option for the below figure

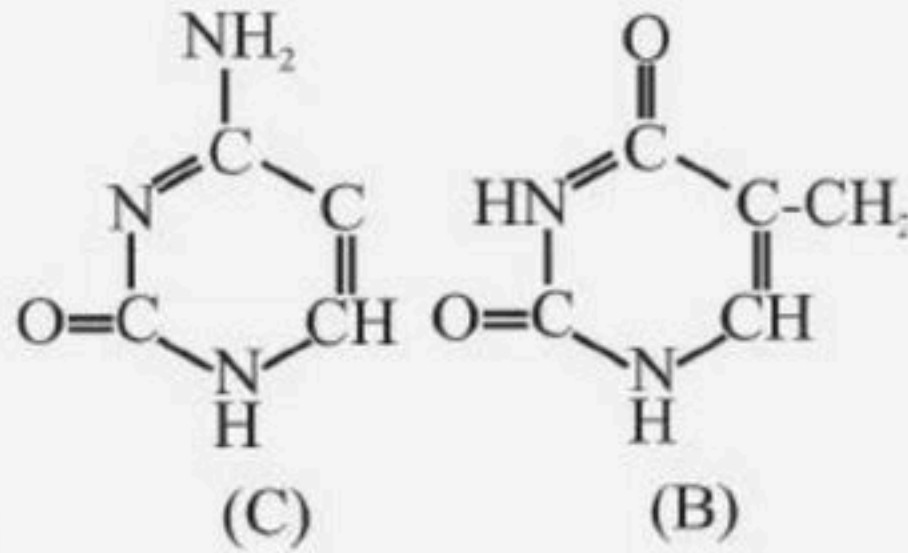
- (1) A - nucleotide; B - nucleoside; C - Phosphate
- ✓ (2) A - nucleoside; B - nucleotide; C - Phosphate
- (3) A - Phosphate; B - nucleotide; C - nucleoside
- (4) A - nucleoside; B - Phosphate; C - nucleotide



37. Identify the purine out of the structure.



(A) → Adenine



(C)

(B)

(1) B – Adenine

✓ (2) A – Adenine double Ring

(3) C – Adenine

(4) A – Guanine



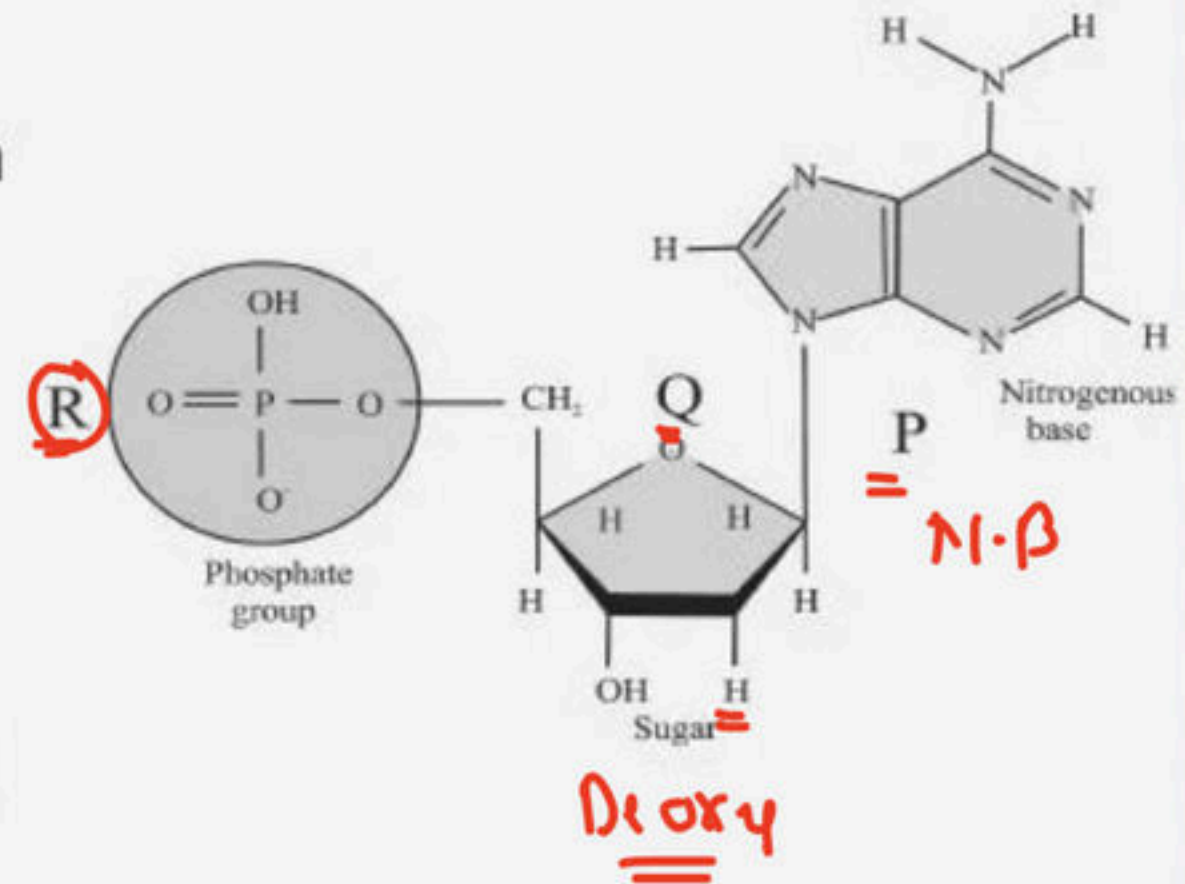
38. Identify .

(1) P - Phosphate group, Q-Deoxyribose sugar, R-Nitrogen base

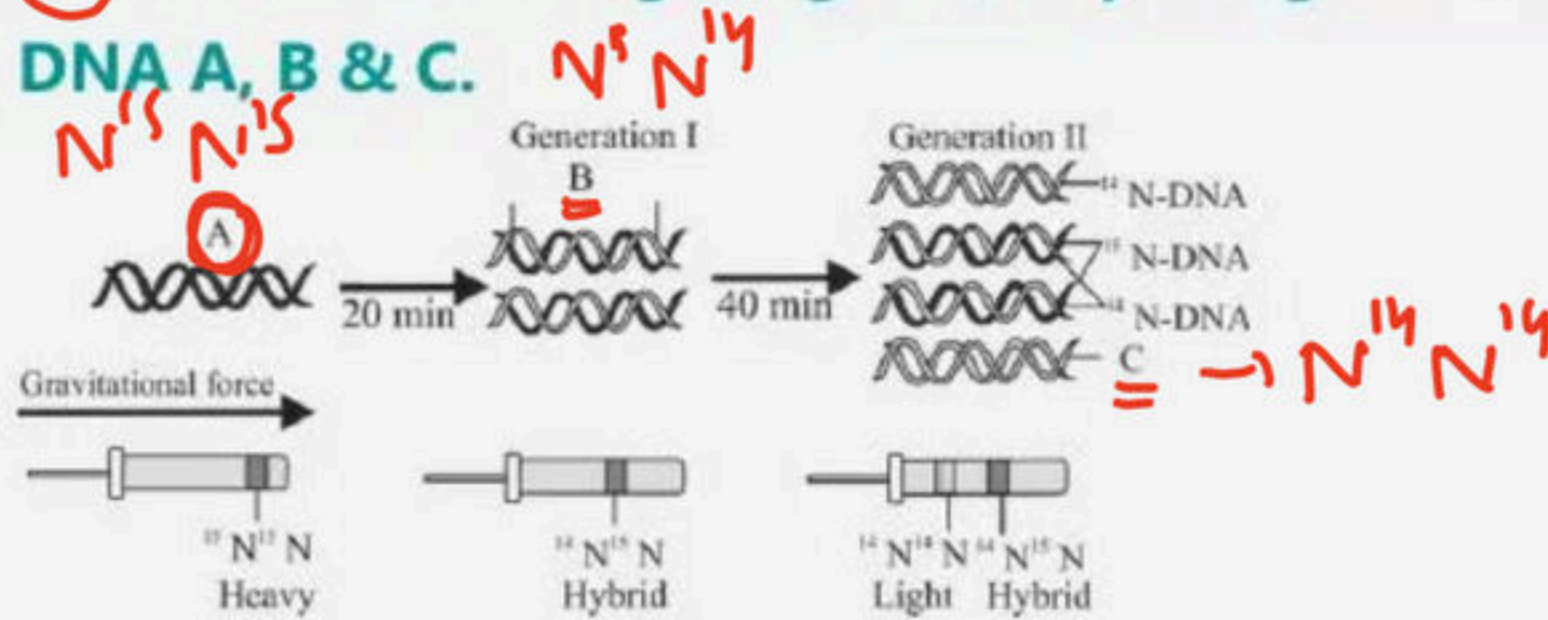
✓ (2) P-Nitrogen base, Q-Deoxyribose sugar, R-Phosphate group

(3) P-Nitrogen base, Q-Ribose sugar, R-Phosphate group

(4) P-Phosphate group, Q-Ribose sugar, R-Nitrogen base



39. In the following diagram, depicting the Messelson & Stahl expt. Identify the type of DNA A, B & C.



(Separation of DNA by centrifugation)

(1) A – N¹⁴ N¹⁴, B – N¹⁵ N¹⁴, C – N¹⁵ N¹⁵

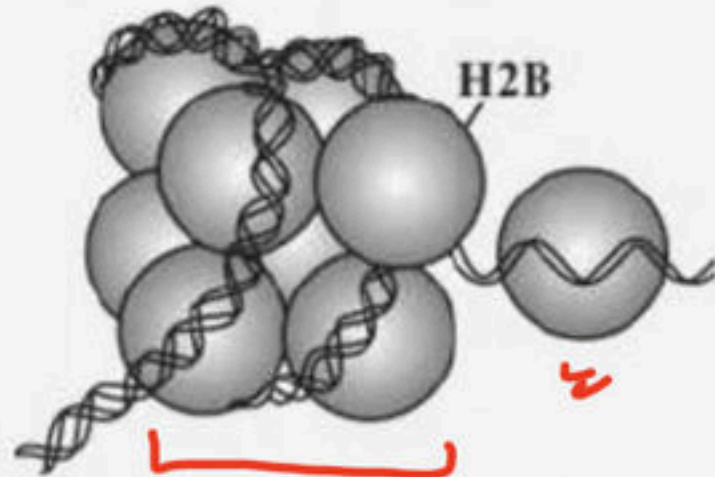
✓ (2) A – N¹⁵ N¹⁵, B – N¹⁵ N¹⁴, C – N¹⁴ N¹⁴

(3) A – N¹⁴ N¹⁴, B – N¹⁴ N¹⁴, C – N¹⁵ N¹⁵

(4) A – N¹⁵ N¹⁵, B – N¹⁵ N¹⁵, C – N¹⁴ N¹⁴



40. In the diagram how many histone proteins are depicted



$$8 + 1 = 9$$

✓ (1) 9

(2) 8

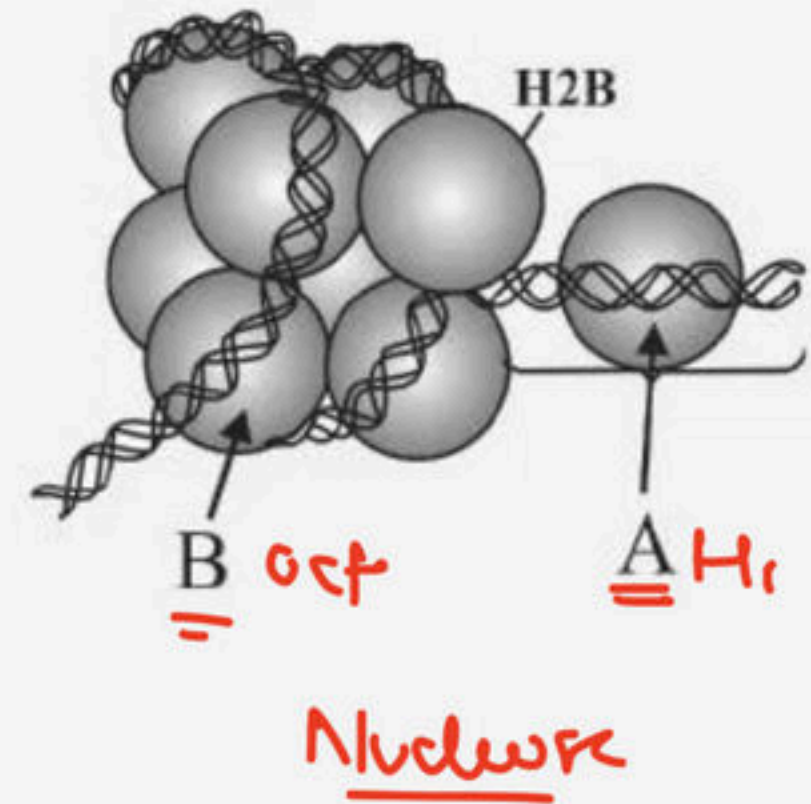
(3) 10

(4) 11



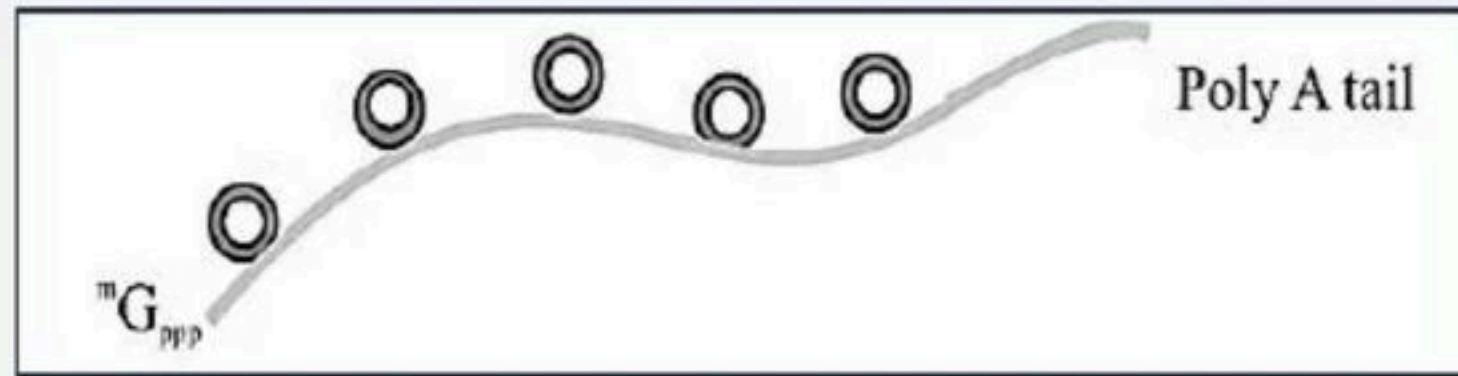
41. Identify the structure & labelled parts A, B.

- | | |
|----------------------------|-------------|
| (1) Nucleus – A – H1 | B – Octamer |
| (2) Nucleus – A – Octamer | B – H1 |
| ✓ (3) Nucleosome – A – H1 | B – Octamer |
| (4) Nucleosome – A Octamer | B – H1 |



42. Identify the process showed in the process diagram

- (1) Capping
- (2) RNA splicing
- (3) Poly adenylation
- (4) ☒ All of these



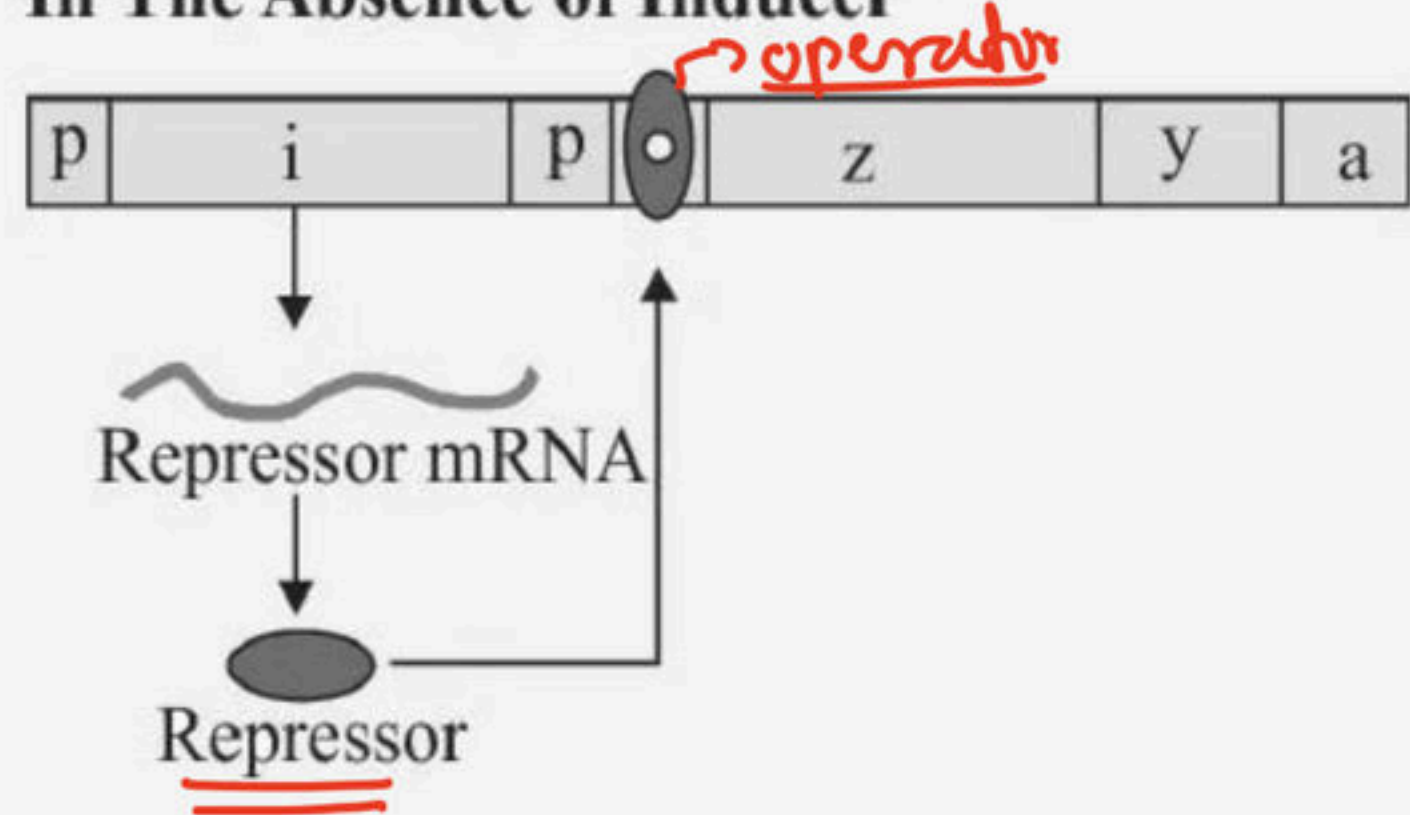
Post transcriptional change



43. Identify the diagram.

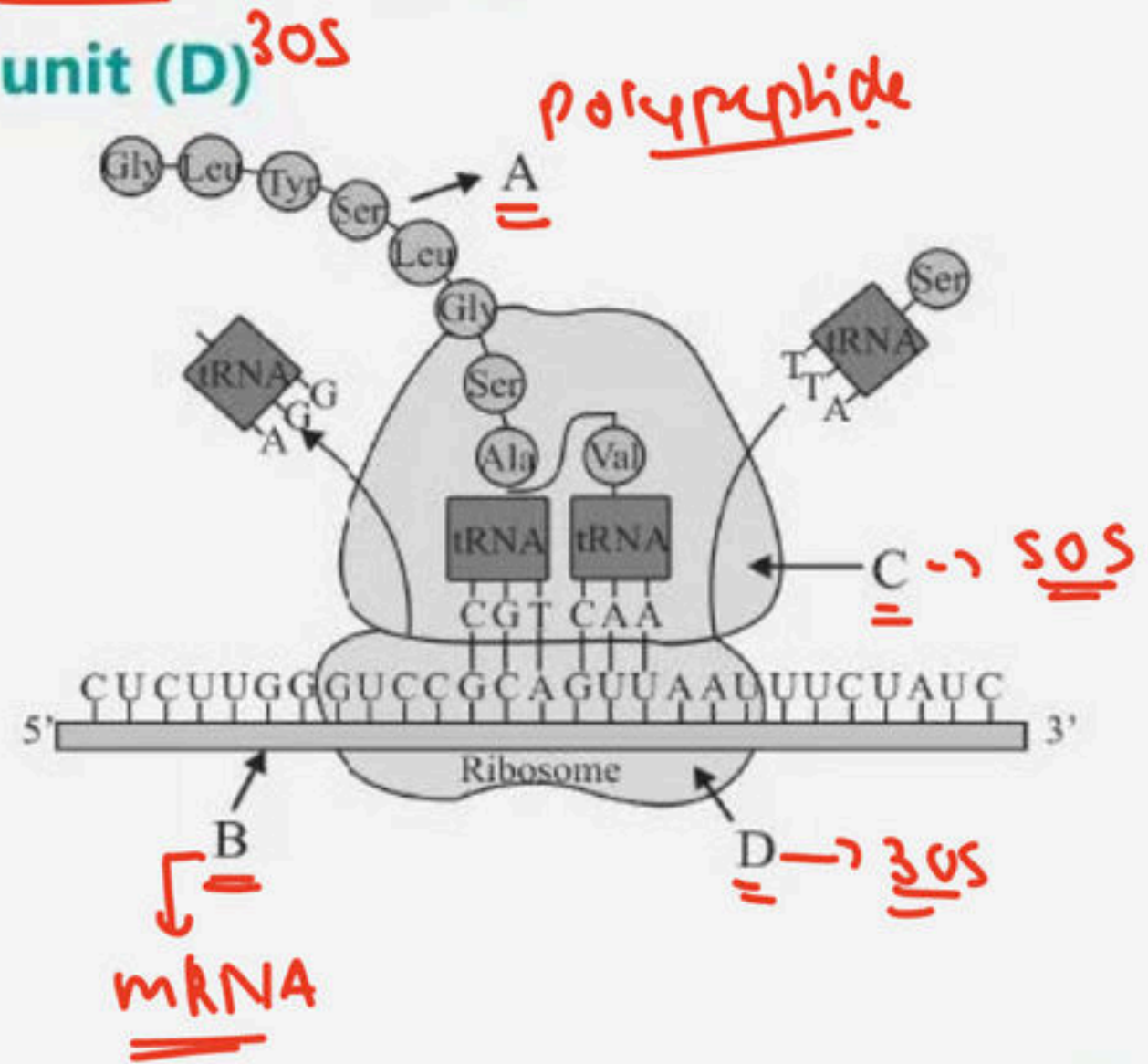
- (1) Lac operon (switched on)
- ✓ (2) Lac operon (switched off)
- (3) Try operon (switched on)
- (4) Trp operon (switched off)

In The Absence of Inducer



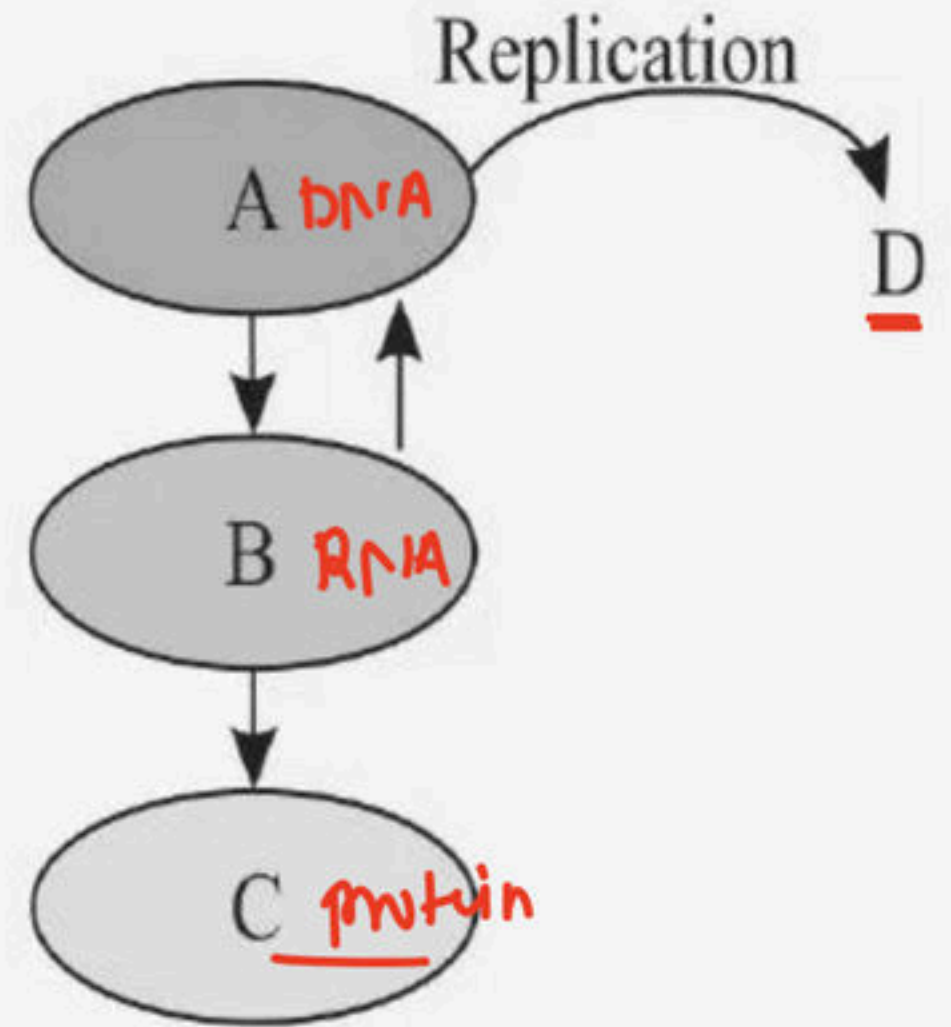
44. Identify the structure that contains codon (B) chain of amino acid, (A) large (C) small ribosomal unit (D) ribosomal unit in a prokaryote

- ✓ (1) m-RNA, polypeptide, 50 S, 30 S
- (2) r-RNA, polypeptide, 50 S, 30 S
- (3) m-RNA, polypeptide, 60 S, 40 S
- (4) m-RNA, polypeptide, 60 S, 40 S



45. The diagram represents the central dogma of molecular biology. Choose the correct combination of labelling.

- (1) A-DNA B-RNA C-Protein D-DNA
- (2) A-RNA B-DNA C-Protein D-RNA
- (3) A-DNA B-Protein C-Protein D-DNA
- (4) A-Protein B-DNA C-DNA D-Protein



46. Match the columns I and II.

Column I		Column II	
A	DNA-dependent DNA polymerase	P	Replication is continuous
B	Deoxyribonucleoside triphosphates	Q	Average rate of polymerisation <u>2000</u> bp per second
C	Polarity <u>3' → 5'</u> of parental strand	R	Dual prupose
D	Polarity 5' → 3' of oarental strand	S	Replication is discontinuous

(1) A – R, B – P, C – Q, D – S

✓ (2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



47. Match the columns I and II.

Column I		Column II	
A	Promoter →	P	Provides binding site for DNA
B	Terminator	Q	Flanked portion
C	Structural gene ✗	R	End of the <u>process</u> of transcription

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

✓ (3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



48. Match the columns I and II WRT Translation.

Column I		Column II	
A	Ribosome	P	Recognised <u>only</u> by the initiator tRNA
B	Release factor	Q	Moves from codon to codon
C	Start codon	R	Binds to the <u>stop codon</u>
D	UTRs	S	Present at the <u>both 5' -end and 3' -end</u>

(1) A – R, B – P, C – Q, D – S

✓ (2) A – Q, B – R, C – P, D – S

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



49. Match Column- I with Column- II

Choose the correct answer from the options given below:

Column I		Column II	
A	Gene 'a'	P	β -galactosidase
B	Gene 'y'	Q	<u>Transacetylase</u>
C	Gene 'i'	R	Permease
D	Gene 'z'	S	Repressor protein

202 S

(1) A-Q, B-P, C-S, D-R

✓ (2) A-Q, B-R, C-S, D-P

(3) A-R, B-S, C-P, D-Q

(4) A-R, B-P, C-S, D-Q



50. Match the columns I and II.

Column I		Column II	
A	Friedrich Meischer	P	Purine: pyrimidine ratio equal one
B	James Watson & Francis Crick	Q	Nuclein
C	Erwin Chargaff	R	1953

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



51. Match the columns I and II wrt HGP.

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Column I		Column II	
A	<i>Caenorhabditis elegans</i>	P	Plant
B	<i>Arabidopsis</i>	Q	Genes that are expressed as RNA
C	EST	R	A free living <u>non-pathogenic nematode</u>
D	Sequence annotation	S	Assigning different regions in the sequence with functions

- ✓ (1) A – R, B – P, C – Q, D – S
- (2) A – Q, B – R, C – P, D – S
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



52. Match the columns I and II wrt HGP.

Column I		Column II	
A	Human genome	P	30000 bases
B	Average gene	Q	<u>3164.7</u> million nucleotide
C	Chromosome I	R	2968 genes
D	Y chromosome	S	231 genes

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- (1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



53. Match the columns I and II.

Column I		Column II	
A	Exons	P	Absent in RNA
B	Introns	Q	Functional unit of inheritance
C	Gene	R	Processed RNA

- ☒ (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



54. Match the columns I and II with respect to DNA.

Column I

Column II

A Base pairing

P Sugar of DNA

B Deoxyribose

Q Unique property

^{pur.}
C Thymine

R Nitrogen base of DNA

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

☒ (4) A – Q, B – P, C – R



55. Match the columns I and II.

Column I

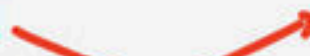
Column II

A Capping



P 5' -end of hnRNA

B Splicing



Q 3' -end of hnRNA

C Tailing



R Removal of introns

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

✓ (3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

V. Imp



56. Match the columns I and II.

Column I		Column II	
A	DNA-dependent DNA polymerase	P	Joining of discontinuously synthesized fragments
B	DNA ligase	Q	Polymerisation only in one direction $5' \rightarrow 3'$
C	Origin of replication	R	Initiate replication

- (1) A – R, B – P, C – Q
- (2) A – Q, B – R, C – P
- (3) A – P, B – R, C – Q
- (4) A – Q, B – P, C – R



57. Match the columns I and II.

Column I		Column II	
A	Process of making <u>RNA</u> from <u>DNA</u>	P	Human genome project
B	Process of protein synthesis	Q	Transcription
C	Determination of complete nucleotide sequence	R	Translation

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



58. Match the columns I and II.

Column I		Column II	
A	Unambiguous	P	Some amino acids are coded by more than one codon
B	Degenerate	Q	AUG
C	Dual functions	R	One codon codes for <u>only one amino acid</u>
D	Universal	S	From bacteria to human UUU would code for phenylalanine

- (1) A – R, B – P, C – Q, D – S
- (2) A – Q, B – R, C – P, D – S
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



59. Match the columns I and II.

Column I		Column II	
A	AUG	P	Phenylalanine
B	UUU	Q	Methionine
C	UUC	R	Phenylalanine
D	UAG	S	Non-sense codon

2025

- (1) A – R , B – P , C – Q, D – S
(2) A – Q , B – R , C – S, D – P
(3) A – P , B – R , C – Q, D – S
(4) A – Q , B – P , C – R , D – S



60. Match the columns I and II.

Column I		Column II	
A	mRNA messenger RNA	P	In ribosomes in cytoplasm
B	tRNA Transfer RNA	Q	From <u>nucleus</u> to <u>cytoplasm</u>
C	rRNA ribosomal RNA	R	In cytoplasm

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R

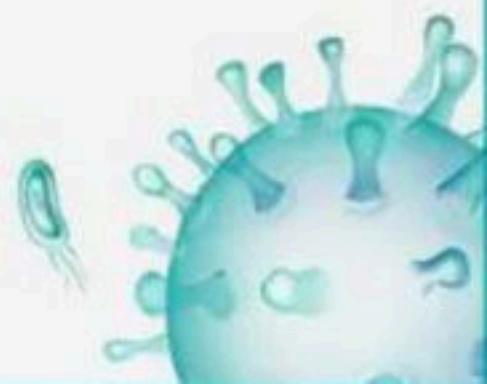


61. The codon on mRNA are:

CAU-CCU-AAA-CUG.

Identify the correct sequence of amino acids.

- ✓ (1) His-Pro-Lys-Leu
- (2) Pro-His-Lys-Leu
- (3) His-Pro-Leu-Lys
- (4) Pro-Leu-Lys-His



62. Match the columns I and II.

Column I		Column II	
A	Repressor	P	Increases permeability of lactose
B	Betagalactosidase	Q	Hydrolysis of the disaccharide
C	Permease	R	Acetylation of galactosides
D	Transacetylase	S	Binds to the operator

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- (1) A – R, B – P, C – Q, D – S
- ✓ (2) A – S, B – Q, C – P, D – R
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



63. Match the columns I and II.

Column I		Column II	
A	Frederick Griffith →	P	<i>Streptococcus pneumoniae</i>
B	Alfred Hershey & Matha Chase	Q	<i>E.coli</i>
C	Mathew Messelson & Franklin Stahl X	R	Bacteriophages
D	Taylor →	S	<i>Vicia faba</i>

- (1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
✓ (3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



64. Match the columns I and II.

Column I		Column II	
A	mRNA messenger RNA	P	Structural & catalytic role
B	tRNA- Transfer RNA	Q	Template for <u>translation</u>
C	rRNA- ribosomal RNA	R	Brings <u>aminoacids</u> and reads the <u>genetic code</u>

(1) A – R, B – P, C – Q

☒ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



65. Match the columns I and II.

Column I		Column II	
A	Cistron	P	Eukaryotes
B	Monocistronic	Q	DNA coding for a <u>polypeptide</u>
C	Polycistronic	R	Bacteria

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- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



66. Match the columns I and II.

Column I		Column II	
A	Promoter 	P	<u>5'</u> -end (upstream) of the structural gene
B	Terminator 	Q	Upstream or downstream to the promoter
C	Additional regulatory sequences 	R	3' -end (downstream) of the coding strand

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

 (3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



67. Match the columns I and II.

Column I		Column II	
A	Replication fork	P	Polyploidy
B	Replication of DNA	Q	Small opening of the <u>DNA helix</u>
C	Failure in cell division after DNA replication	R	S-phase of the cell cycle

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



68. Match the columns I and II wrt tRNA.

Column I		Column II	
A	Anticodon loop	P	It binds to specific amino acids
B	Amino acid acceptor end	Q	Clover-leaf
C	Secondary structure	R	<u>Complementary</u>
D	Actual structure	S	<u>Inverted L</u>

- ✓ (1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



69. Match the columns I and II.

Column I		Column II	
A	N-glycosidic linkage	P	Linkage of two nucleotides
B	Phosphoester linkage	Q	Nitrogenous base is linked to the <u>C-1</u> of pentose sugar
C	Phosphodiester bonds	R	Phosphate group is <u>linked</u> to the <u>OH group</u> of the <u>5-C</u>
D	2 Hydrogen bonds	S	Between C & G
E	3 Hydrogen bonds	T	Between A & T

(1) A – R, B – P, C – Q, D – T, E – S

☒ (2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

(4) A – Q, B – T, C – P, D – R, E – S



70. Match the scientists of Column I with their contributions in Column II

Column I		Column II	
A	Griffith	P	Lac operon
B	Jacob & Monod	Q	DNA is the genetic material
C	Meselson & Stahl	R	Transforming principle
D	Hershey and Chase	S	DNA replicates <u>semi-conservatively</u> .

(1) A - P; B - Q; C - R; D - S

(2) A - P; B - S; C - Q; D - R

☒ (3) A - R; B - P; C - S; D - Q

(4) A - R; B - Q; C - P; D - S



71. Match the columns I and II with respect to nucleosome.

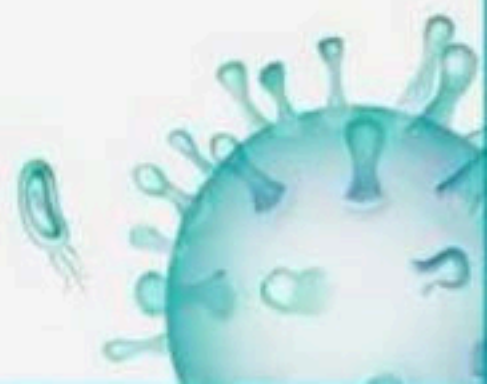
Column I		Column II	
A	Typical nucleosome	P	Negatively charged
B	Histones	Q	200 bp
C	DNA	R	Basic amino acid <u>lysine</u> and arginines

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



72. Match the columns I and II.

Column I		Column II	
A	Bacteriophage $\phi \times 174$ →	P	5386 nucleotides
B	Bacteriophage lambda →	Q	4.6×10^6 bp
C	<i>Escherichia coli</i> →	R	48502 base pairs
D	Human DNA haploid →	S	3.3×10^9 bp

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(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

✓ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



73. Match the columns I and II.

Column I		Column II	
A	Frederick Griffith	P	1952
B	Oswald Avery	Q	British bacteriologist - 1928
C	Alfred Hershey & Martha Chase	R	Radioactive thymidine
D	Taylor	S	1958
E	Mathew Messelson & Franklin Stahl	T	1933 - 34

(1) A – R, B – P, C – Q, D – T, E – S

(2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

✓ (4) A – Q, B – T, C – P, D – R, E – S



74. Match the following:

Column I		Column II	
A	t-RNA	1	Linking of amino-acids
B	m-RNA	2	Transfer of genetic information
C	r-RNA	3	Nucleolar organising region
D	Peptidyl	4	Transfer of <u>amino acid</u> from transferase <u>cytoplasm</u> of <u>ribosome</u>

✓ (1) A-4, B-2, C-3, D-1

(2) A-1, B-4, C-3, D-2

(3) A-1, B-2, C-3, D-4

(4) A-1, B-3, C-2, D-4



75. Match the columns I and II.

Column I		Column II	
A	RNA polymerase I	P	Transcribes snRNAs
B	RNA polymerase II	Q	Transcribes rRNAs
C	RNA polymerase III	R	Transcribes hnRNA

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- (1) A – R, B – P, C – Q
✓ (2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



76. Match the following.

Column I		Column II	
A	VNTR	P	Largest gene
B	Introns and exons	Q	DNA fingerprinting
C	Dystrophin	R	Bulk DNA
D	Satellite DNA	S	Splicing

- (1) A - Q; B - S; C - P; D - R
- (2) A - S; B - P; C - Q; D - R
- (3) A - R; B - S; C - P; D - Q
- (4) A - Q; B - P; C - S; D - R



77. Match the columns I and II.

Column I		Column II	
A	Taylor & colleagues	P	Protein coat
B	Cesium chloride density gradient centrifugation	Q	<u>1958</u> , <u>Visia faba</u>
C	Radioactive sulphur (^{35}S)	R	Semi-conservative replication of DNA

- (1) A – R, B – P, C – Q
✓ (2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



78. Match Column I with Column II

Choose the correct answer from the options given below:

Column I		Column II	
A	Frederick Griffith	P	Genetic code
B	Francois Jacob & Jacque Monod	Q	Semi-conservative mode of DNA replication
C	Har Gobind Khorana	R	Transformation
D	Meselson & Stahl	S	Lac operon

(1) A - R, B - Q, C - P, D - S

✓ (2) A - R, B - S, C - P, D - Q

(3) A - Q, B - R, C - S, D - P

(4) A - S, B - P, C - Q, D - R



79. Match the columns I and II.

Column I		Column II	
A	<i>E.coli</i> DNA lengths →	P	1.36 mm
B	Human DNA length ↘	Q	1 μm
C	<i>E.coli</i> area	R	2.2 m
D	Human nucleus area	S	5 μm

(1) A – R, B – P, C – Q, D – S

(2) A – Q, B – R, C – P, D – S

☒ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



80. Match the columns I and II.

Column I		Column II	
A	Non-histone chromosomal (NHC) proteins	P	Loosely packed chromatin
B	Euchromatin	Q	Higher level packaging of chromatin
C	Heterochromatin	R	Transcriptionally inactive

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(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

✓ (4) A – Q, B – P, C – R



81. Match Column I with Column II:

Choose the correct answer from the options given below :

Column I		Column II	
A	RNA polymerase III	P	snRNPs
B	Termination of transcription	Q	Promotor
C	Splicing of Exons	R	Rho factor
D	TATA box	S	<u>SnRNAs</u> , <u>tRNA</u>

- (1) A - Q, B - S, C - P, D - R
(2) A - R, B - Q, C - S, D - P
(3) A - R, B - S, C - P, D - Q
(4) A - S, B - R, C - P, D - Q



82. Match the columns I and II

Column I		Column II	
A	i gene	P	Transacetylase
B	z gene	Q	Repressor of the <u>lac operon</u>
C	y gene	R	Permease
D	a gene	S	Betagalactosidase

v.v. imp

- (1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – S, C – R, D – P



83. Match the columns I and II.

Column I		Column II	
A	H1 Histones	P	Inner core of nucleosom
B	H3 Histones	Q	Connects two nucleosome
C	Linker DNA	R	Present once per <u>200</u> base pairs
D	Human nucleus area	S	5 μm

- ✓ (1) A – R, B – P, C – Q, D – S
- (2) A – Q, B – R, C – P, D – S
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



84. Match the columns I and II.

Column I		Column II	
A	RNA	P	Expression
B	Mendelian characters	Q	Nitrogen base of RNA
C	Uracil	R	QB bacteriophage

✓ (1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



85. Match the columns I and II.

Column I		Column II	
A	George Gamow →	P	Physicist
B	Har Gobind Khorana	Q	Cell-free system for protein synthesis
C	Marshall Nirenberg	R	Homopolymers & copolymers
D	Severo Ochoa →	S	Polynucleotide phosphorylase

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- (1) A – R, B – P, C – Q, D – S
 (2) A – Q, B – R, C – P, D – S
 ✓ (3) A – P, B – R, C – Q, D – S
 (4) A – Q, B – P, C – R, D – S



86. Match the columns I and II.

Column I		Column II	
A	Friedrich Meischer	P	X-ray diffraction data
B	Maurice Wilkins & Rosalind Franklin	Q	Thymine
C	5-methyl uracil	R	1869

- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



87. Match the columns I and II wrt HGP.

Column I		Column II	
A	1990	P	Development of a new area in biology
B	2003	Q	Address the ethical, legal and social issues
C	ELSI	R	HGP was completed
D	Bioinformatics	S	HGP was launched

2021

- (1) A – R, B – P, C – Q, D – S
(2) A – Q, B – R, C – P, D – S
☒ (3) A – S, B – R, C – Q, D – P
(4) A – Q, B – P, C – R, D – S



88. Match the processes of Column I with the enzyme involved in Column II and select the correct options among the following

Select the code for the correct answer from the options given below

Column I		Column II	
A	DNA replication	P	RNA polymerase
B	Translation	Q	<u>DNA polymerase</u>
C	Transcription	R	Reverse transcriptase
D	Reverse transcription	S	Aminoacyl synthetase

(1) A - Q; B - S; C - R; D - P

✓ (2) A - Q; B - S; C - P; D - R

(3) A - Q; B - R; C - S; D - P

(4) A - Q; B - S; C - R; D - P



89. Statement A : DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869.

Statement B : The base pairing confers a very unique property to the polynucleotide chains.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct.

(4) Both Statements are incorrect.



90. Matthew Meselson and Franklin Stahl grew ^{E. coli} in a medium containing $^{15}\text{NH}_4\text{Cl}$ (^{15}N is the heavy isotope of nitrogen) as the only nitrogen source for many generations. The result was that N^{15} was incorporated into newly synthesised DNA (as well as other nitrogen containing compounds).

- ✓ (1) E. coli, heavy, ^{15}N
- (2) bacteriophage, heavy, ^{15}N
- (3) E. coli, radioactive, ^{15}N
- (4) bacteriophage, radioactive, ^{15}N



91. Sequences prior to the start point of transcription are called

(1) Non coding sequences

✓ (2) Upstream sequences

(3) Coding sequences

(4) Downstream sequences



92. Statement A : Griffith concluded that the ^Sstrain bacteria had somehow been transformed by the heatkilled R strain bacteria. ✗

Statement B : Some 'transforming principle', transferred from the heatkilled S strain, had enabled the R strain to synthesise a smooth ✗
polysacch
polypeptide coat and become virulent.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- ✓ (4) Both Statements are incorrect.



✓ 93. Statement A : A single-strand of mRNA is capable of forming different polypeptide chains in prokaryotes.

✓ Statement B : Termination codons occur in polypeptide. mRNA

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



94. The single stranded DNA of phage $\phi \times 174$ has 5400 nucleotides. If average protein contains 200 amino acids, how many different proteins could this phage

DNA code ?

(1) 100

(2) 90

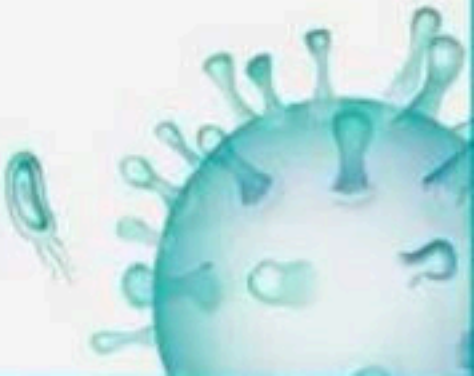
✓ (3) 9

(4) 27

$$= \frac{5400}{3} = \underline{1800 \text{ codon}} \rightarrow$$

$$\underline{200 \text{ a.a.}} = \underline{200 \text{ codon}}$$

$$\frac{1800}{200} = 9$$



95. In prokaryotes, DNA is held together by which proteins?

- (1) Negatively charged proteins
- (2) Non-histone chromosomal proteins
- ✓ (3) Positively charged proteins
- (4) Histone proteins



96. If both the strands copied during transcription, then what will happen?

(1) The segment of DNA would be coding for two different proteins ✓

(2) Two RNA will be produced simultaneously complementary to each other. ✓

(3) There will be formation of double helical RNA ✓

✓ (4) All of the above



97. Statement A : The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952).

Statement B : They worked with viruses that infect bacteria called bacteriophages.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



98. Name the nucleotide added to 5' end of hnRNA in capping.

- (1) Ethyl guanosine monophosphate
- (2) Ethyl guanosine triphosphate
- ✓ (3) Methyl guanosine triphosphate
- (4) Methyl guanosine monophosphate



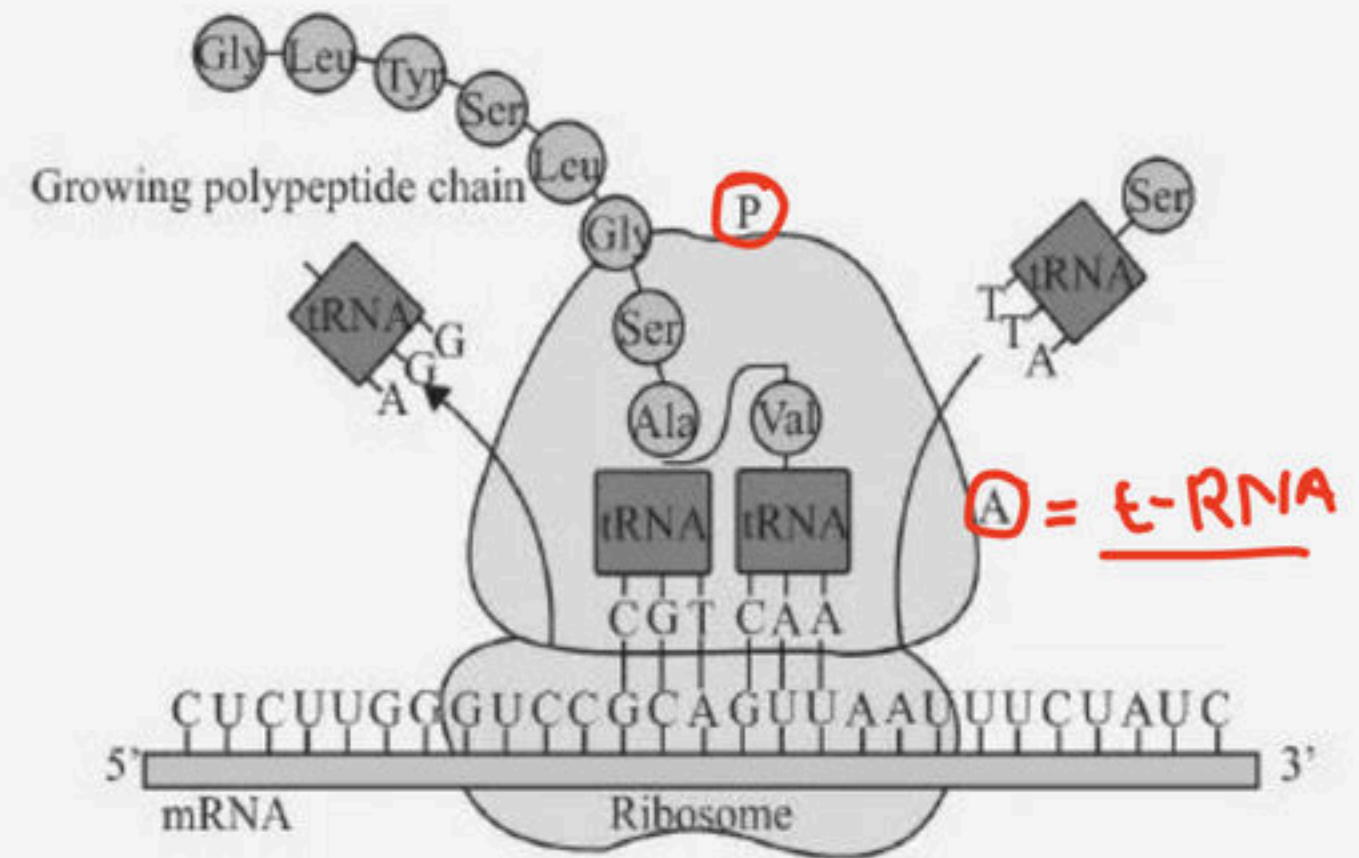
99. When mRNA is synthesized on DNA, the unwanted mRNA regions are removed and regions coding for amino acids are joined together, this process is called as

- ✓ (1) Splicing
- (2) Lysogeny
- (3) Replication
- (4) Central dogma



100. With respect to ribosome what does A & P represent.

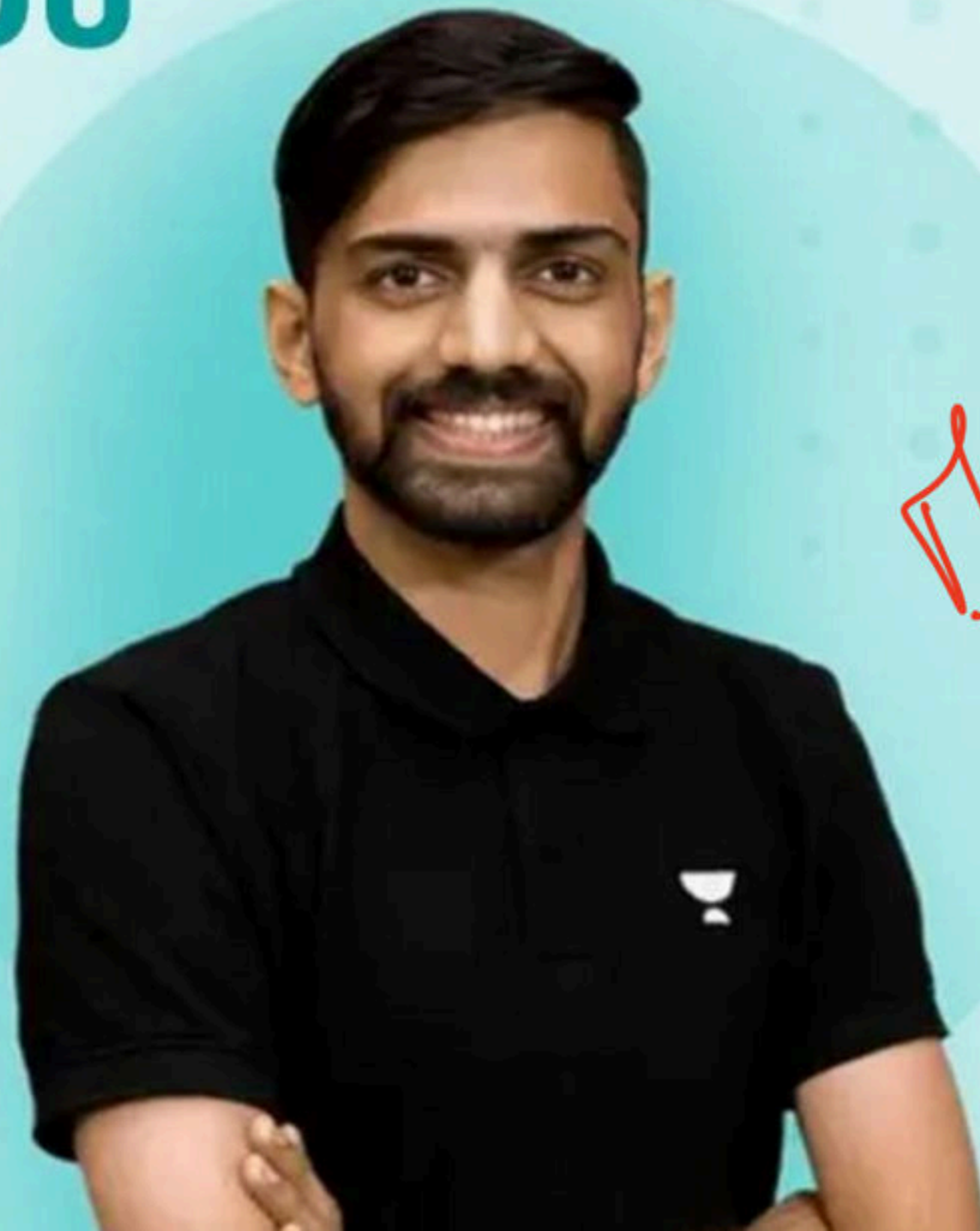
- (1) A - Peptidyl site P - t-RNA binding
- ✓ (2) A - tRNA-binding site P - peptidyl site
- (3) A - m - RNA site P - peptidyle site
- (4) A - t-RNA binding site P - ATP site



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